The weak-weight non-unitary Paley question: endpoint failure and interior validity

Candidate counterexample packet for arXiv:2409.17969

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Outcome. The proposed replacement of strong L^1 control by the weak c-norm is false as stated: it fails at $p = q = 1$ on every rank-one noncompact symmetric space. Nevertheless it is true for every $1 < p < 2$ and every strict interior line $p < q < p'$. For $1 < p < 2$, only the two non-unitary edges $q = p, p'$ remain unresolved.

1. The exact question

Let $\mathbb{X} = G/K$ be a rank-one Riemannian symmetric space of noncompact type, $n = \dim \mathbb{X}$, and

$$\mathrm{d}\mu(\lambda) = |c(\lambda)|^{-2}\,\mathrm{d}\lambda, \qquad \rho_q = (2/q - 1)\rho.$$

For $1 \le q < \infty$ set

$$\mathcal{F}_q f(\lambda) = \left(\int_K |\widetilde{f}(\lambda + i\rho_q, k)|^q \,\mathrm{d}k\right)^{1/q} \frac{|\lambda + i\rho_q|}{|\lambda + i\rho_q + i2\rho|}.$$

Theorem 1.11 of the source assumes $u \in L^1(\mu)$ and, for $1 \le p \le 2$, $p \le q \le p'$, proves

$$\|\mathcal{F}_q f\|_{L^p(u^{2-p}\,\mathrm{d}\mu)} \le C_{p,q} \|u\|_{L^1(\mu)}^{2/p-1} \|f\|_{L^p(\mathbb{X})}. \tag{1}$$

Remark 5.2 asks whether one may instead use

$$\|u\|_{1,\infty;\mu} := \sup_{\alpha>0} \alpha\, \mu\{u > \alpha\} < \infty.$$

Remark 5.2. (1) It is natural to ask whether Theorem 1.11 holds if we assume $\|u\|_{c,\infty}$ is finite, as we did in Theorem 1.9, instead of $\|u\|_{L^1(|c(\lambda)|^{-2})} < \infty$. We note that if $\|u\|_{c,\infty} < \infty$, then we can prove that the operator $\mathcal{S}_{i\rho,0}$, defined in (5.3), is of weak type $(1,1)$ (Theorem 1.9). However, it may not be of strong type $(1,1)$, which seems necessary to apply Stein's analytic interpolation.

Figure 1: The exact question and its endpoint warning, source PDF page 27.

Theorem 1 (Sharp split for the proposed upgrade)**.** *On every rank-one* $\mathbb{X}$*:*

1. *The weak-weight version of* (1) *is false for* $p = q = 1$*. Thus the all-exponent question in Remark 5.2 has a negative answer.*
2. *If* $1 < p < 2$ *and* $p < q < p'$*, then*

$$\|\mathcal{F}_q f\|_{L^p(u^{2-p}\,\mathrm{d}\mu)} \le C_{p,q} \|u\|_{1,\infty;\mu}^{2/p-1} \|f\|_{L^p(\mathbb{X})}. \tag{2}$$

Proof intuition. The endpoint fails at the critical scale. Plancherel balls have volume $\asymp R^n$, so $u(\lambda) = (1+|\lambda|)^{-n}$ is weak L^1 but has logarithmically divergent weighted mass. Shrinking approximate identities have Fourier transform close to one on balls of radius R and detect that divergence. In the strict interior, the source's pointwise restriction and Hausdorff–Young estimates real-interpolate to the sharper Lorentz target $L^{p',p}$; this is exactly the space dual to the power of the weak weight.

2. Endpoint counterexample

The source records the rank-one estimate

$$|c(\lambda)|^{-2} \asymp \lambda^2(1+|\lambda|)^{n-3}. \tag{3}$$

Take

$$u(\lambda) = (1+|\lambda|)^{-n}.$$

For $0<\alpha<1$, the set $\{u>\alpha\}$ is contained in an interval of radius $\alpha^{-1/n}$. Equation (3) gives

$$\mu\{u>\alpha\} \le C\alpha^{-1}, \qquad \text{so} \qquad \|u\|_{1,\infty;\mu}<\infty.$$

On the other hand, again by (3),

$$\int_1^R u(\lambda)\,\mathrm{d}\mu(\lambda) \ge c\int_1^R \lambda^{-n}\lambda^{n-1}\,\mathrm{d}\lambda = c\log R. \tag{4}$$

Choose $f_R \in C_c^\infty(\mathbb{X})$ nonnegative with

$$\|f_R\|_1 = 1, \qquad \operatorname{supp} f_R \subset B(o,R^{-1}).$$

The Iwasawa projection satisfies $|H(x^{-1}k)| \le d(o,x)$. At the upper boundary $q=1$,

$$\widetilde{f}_R(\lambda+i\rho,k) = \int_{\mathbb{X}} f_R(x)e^{(i\lambda-2\rho)H(x^{-1}k)}\,\mathrm{d}x.$$

Using $|e^z-1|\le e^{|z|}|z|$, choose a fixed $b>0$ small enough. Uniformly in k and $|\lambda|\le bR$,

$$|\widetilde{f}_R(\lambda+i\rho,k)-1| \le \tfrac{1}{2} \tag{5}$$

for all sufficiently large R. Moreover $|\lambda+i\rho|/|\lambda+i3\rho|$ is bounded below for $\lambda\ge 1$. Consequently (4) and (5) imply

$$\|\mathcal{F}_1 f_R\|_{L^1(u\,\mathrm{d}\mu)} \ge c\log R,$$

whereas the proposed right side is $C\|u\|_{1,\infty;\mu}\|f_R\|_1$, independent of R. This proves Theorem 1(1).

3. The strict interior is positive

Fix $1<p<2$ and $p<q<p'$. Without the harmless scalar factor, put

$$G_q f(\lambda) = \left(\int_K |\widetilde{f}(\lambda+i\rho_q,k)|^q\,\mathrm{d}k\right)^{1/q}.$$

Choose

$$p<r<\min\{q,q'\}.$$

Then $1<r<2$ and $r<q<r'$. The pointwise restriction estimate reproduced as Theorem 2.3 in the source and its Hausdorff–Young estimate, Theorem 2.2, give respectively

$$G_q : L^1(\mathbb{X}) \longrightarrow L^\infty(\mu), \qquad G_q : L^r(\mathbb{X}) \longrightarrow L^{r'}(\mu).$$

The map G_q is sublinear. Real interpolation with

$$\frac{1}{p} = 1-\theta+\frac{\theta}{r} \quad\Longrightarrow\quad \frac{\theta}{r'} = \frac{1}{p'}$$

therefore yields the Lorentz refinement

$$\|G_q f\|_{L^{p',p}(\mu)} \le C_{p,q}\|f\|_{L^p(\mathbb{X})}. \tag{6}$$

Now $G_q^p \in L^{1/(p-1),1}(\mu)$, while the weak-weight hypothesis gives

$$u^{2-p}\in L^{1/(2-p),\infty}(\mu), \qquad \|u^{2-p}\|_{1/(2-p),\infty} = \|u\|_{1,\infty;\mu}^{2-p}.$$

The two displayed exponents are conjugate. Lorentz Hölder and (6) show

$$\begin{aligned}\int_{\mathbb{R}} G_q f(\lambda)^p u(\lambda)^{2-p}\,\mathrm{d}\mu(\lambda) &\le C\|G_q f\|_{p',p}^p\|u\|_{1,\infty;\mu}^{2-p}\\ &\le C_{p,q}\|f\|_p^p\|u\|_{1,\infty;\mu}^{2-p}.\end{aligned}$$

Finally, the scalar factor in $\mathcal{F}_q$ is at most one because

$$(\rho_q+2\rho)^2-\rho_q^2 = 4\rho(\rho_q+\rho) = \frac{8\rho^2}{q} > 0.$$

Taking pth roots proves (2).

4. Exact remaining boundary

The interpolation choice $p < r < \min\{q, q'\}$ exists exactly when $p < q < p'$. At $q = p$ or $q = p'$ it collapses, and the source's strong Hausdorff–Young target $L^{p'}$ does not by itself imply the needed sharper $L^{p',p}$ estimate. No claim is made for those two edge lines when $1 < p < 2$. The unitary line $q = 2$ is already covered by Theorem 1.9 of the source. Constants in the interior proof necessarily deteriorate as $p \downarrow 1$, consistently with the logarithmic endpoint counterexample.

Upgrade audit. Eight focused stages were completed: exact-scope and literature audits; critical-weight scaling; approximate-identity construction; generalization to all rank-one spaces; interior Lorentz interpolation; weak-weight pairing; and edge-line analysis. The literal question is settled negatively, and its strict interior is settled positively.

Novelty boundary. Bounded primary-source searches through 17 August 2026 found no later paper recording either conclusion. The Lorentz argument is short and may be known folklore; priority requires a broader search and expert review.

References

[1] T. Rana and M. Ruzhansky, *L^p–L^q Fourier multipliers and Hausdorff–Young–Paley inequalities on Riemannian symmetric spaces of noncompact type*, arXiv:2409.17969 (2024).

[2] J. Bergh and J. Löfström, *Interpolation Spaces: An Introduction*, Springer, 1976.

[3] R. O'Neil, *Convolution operators and $L(p, q)$ spaces*, Duke Math. J. 30 (1963), 129–142.